

DataTierAI - Enterprise Data Sources & ETL Foundation

1. Windows File Collection

Definition: Collecting file metadata (name, size, path, type, created date) from a Windows file server for inventory and analysis.

- Q: Why collect Windows metadata? A: To monitor files and storage usage.
- Q: What metadata is collected? A: File name, size, type, path, dates.

2. Linux File Collection

Definition: Collecting metadata from Linux directories such as /home, /var and /etc for enterprise storage analysis.

- Q: Why collect Linux metadata? A: To analyze Linux storage and files.
- Q: Which command is commonly used? A: ls, find (in real systems).

3. Database Metadata Extraction

Definition: Reading information from databases (tables, records, columns) instead of scanning files.

- Q: Why use a database? A: Structured enterprise data.
- Q: Which database was used? A: SQLite.

4. REST API Basics

Definition: REST APIs allow applications to exchange data over HTTP using JSON.

- Q: Common methods? A: GET, POST, PUT, DELETE.
- Q: Response format? A: JSON.

5. GitHub API

Definition: GitHub provides REST APIs to access repository information.

- Q: Why GitHub API? A: Real enterprise metadata source.
- Q: What data collected? A: Repository name, stars, forks, language, etc.

6. API Authentication (PAT)

Definition: Personal Access Token securely authenticates API requests.

- Q: Why PAT? A: Increases rate limit and provides secure access.
- Q: Anonymous vs PAT? A: 60 vs ~5000 requests/hour.

7. Pagination

Definition: Large API responses are split into pages.

- Q: What is per_page? A: Number of records per request.
- Q: Why pagination? A: Efficient transfer of large datasets.

8. JSON Parsing

Definition: Converting JSON responses into Python dictionaries/lists.

- Q: Which function? A: `response.json()`.

9. DataFrame Creation

Definition: Storing structured data in pandas DataFrame for analysis.

- Q: Why DataFrame? A: Easy filtering, cleaning and export.

10. CSV Generation

Definition: Exporting processed data into CSV files for ETL and analytics.

- Q: Which function? A: `df.to_csv()`.